A Micro-turbojet Test Rig To Analyze The Impact Jet Fuels on Emissions and Performance

Siddharth Sharma[1,], Daniel Lieder[2], Simone Saini[1], Anand Kumar[1], Patrick Kirchen[1], Jan Goeing[1,2], Steven Rogak[1]*

[1]Department of Mechanical Engineering University of British Columbia
[2]Institute of Jet Propulsion and Turbomachinery, Technische Universität Braunschweig

Motivation: Aviation's Climate Impact

- Aviation contributes **3-5%** to global anthropogenic climate forcing [1,2].

- **More than half** of aviation's **total climate impact arises from non-CO_2 effects** [1,3,4].

Non-CO_2 Forcing Mechanisms

- Reactive nitrogen species drive near-field chemical transformations and interact with the atmosphere [5,6].

- Emitted soot can act as Ice Nucleating Particles (INPs), **forming contrails** & influence optical properties of clouds, further amplifies radiative warming [3,7].

Because multiple coupled processes contribute to net climate forcing, CO_2 alone is an incomplete proxy for aviation's total impact.

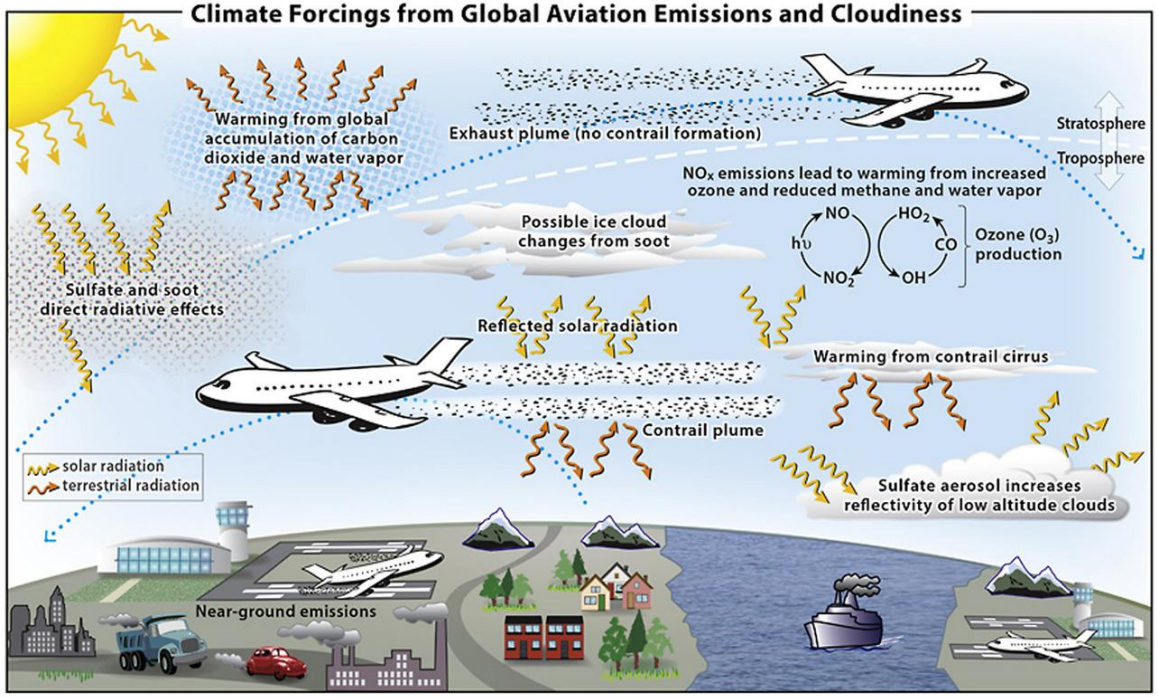


Illustration from [1].

[1] Lee et al., *Atmos. Environ.*, 2021 | [2] Gössling and Humpe, *Glob. Environ. Change*, 2020 | [3] Bickel et al., *J. Climate*, 2020 | [4] Teoh et al., *Atmos. Chem. Phys.*, 2022 | [5] Mishra and Chandel, *Int. J. Turbo Jet Engines*, 2019 | [6] Blakey et al., *Proc. Combust. Inst.*, 2011 | [7] Corporan et al., *Energy Fuels*, 2011

Motivation: How Can We Decrease Non-CO2 Emissions

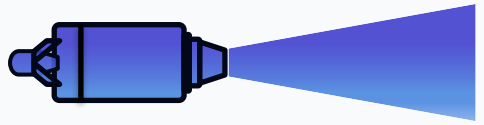
Industry Shift & Alternative Fuels

ICAO Regulations

ICAO has introduced standards for non-volatile PM (CAEP 11) and Annex 16 specifies limits for key gaseous non-CO_2 species [8].

The Interest in SAFs

Sustainable Aviation Fuels (SAFs) are positioned as a practical near-term pathway to reduce climate impact without altering existing airframes.



Emissions Data

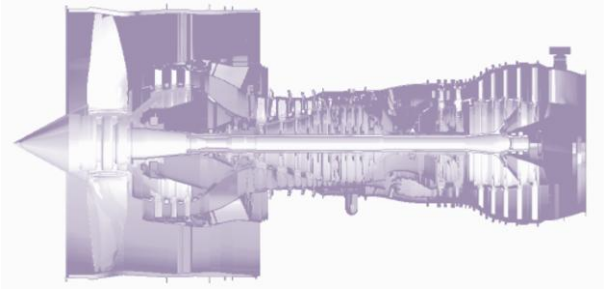
To characterize the emissions of these alternative fuels, we need to gather data on how they burn. But many promising SAF pathways are not yet certified for commercial use.

[8] (ICAO Annex 16)

Motivation: Characterizing Emissions

Pollutant formation and resultant climate impact are highly sensitive to fuel chemistry, and operating point dependencies.
We need a well instrumented laboratory platform to examine these sensitivities.

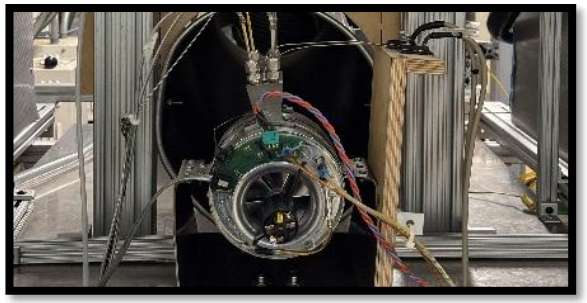
The Challenges



Characterization of SAFs using a full-scale turbofan is:

- **Highly Regulated**
- **Requiring tens of thousands of liters of fuel**
- **Very Expensive**

1. Micro-Turbojet (MTJ) Test Rig



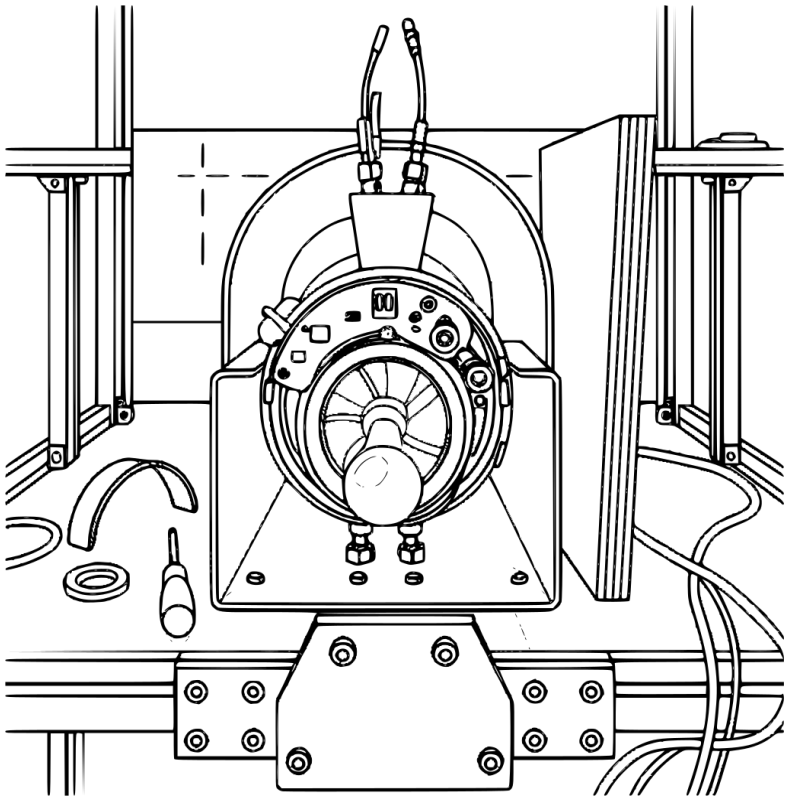
Why? Cost-effective, controlled platform:

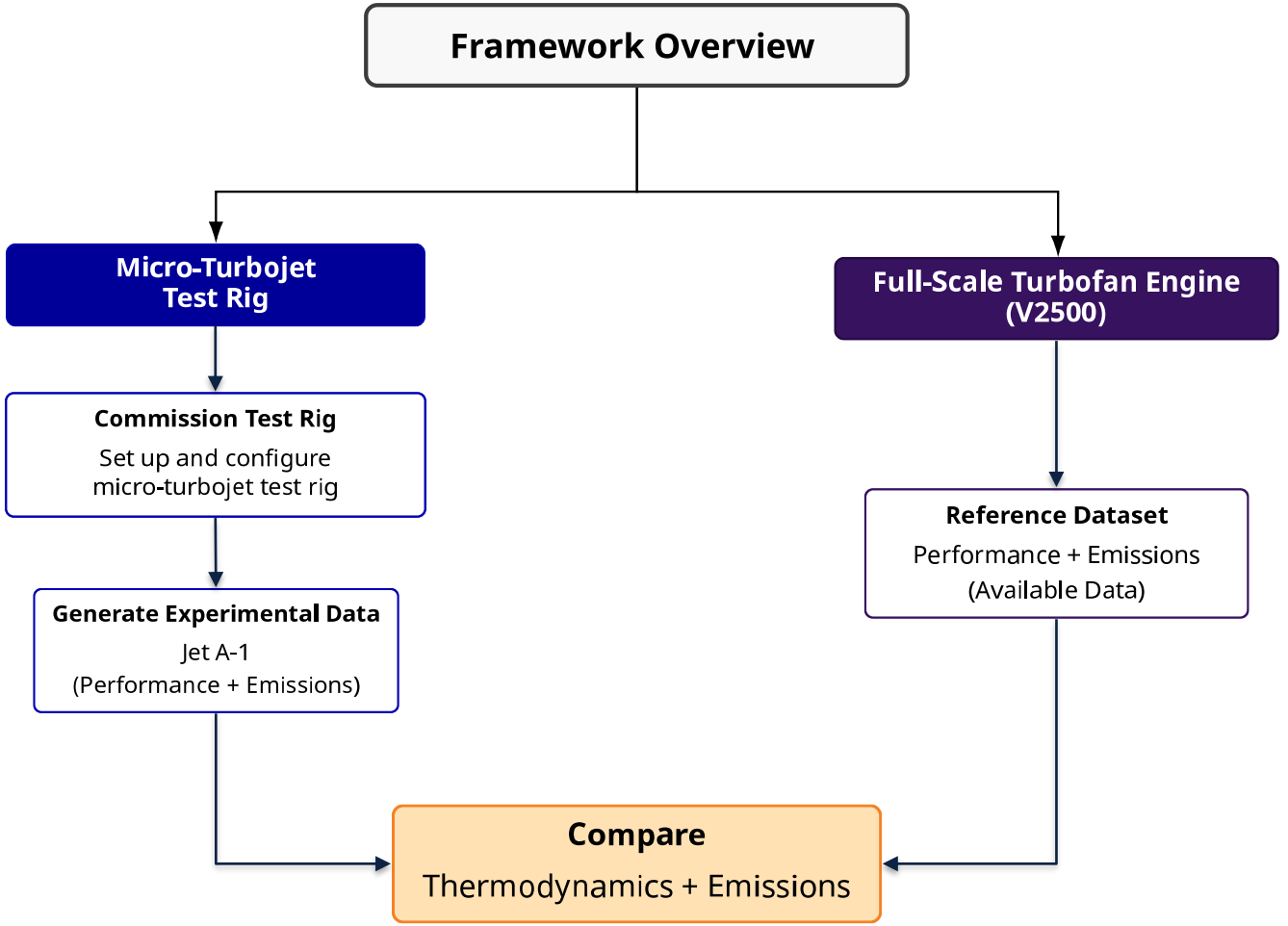
- ✓ **Repeatable experiments**
- ✓ **Modest fuel use across operating envelope**
- ✓ **Enables testing of uncertified fuels**

V2500 image adapted from, [9] Göing, *PhD thesis*, 2023

Research Objectives

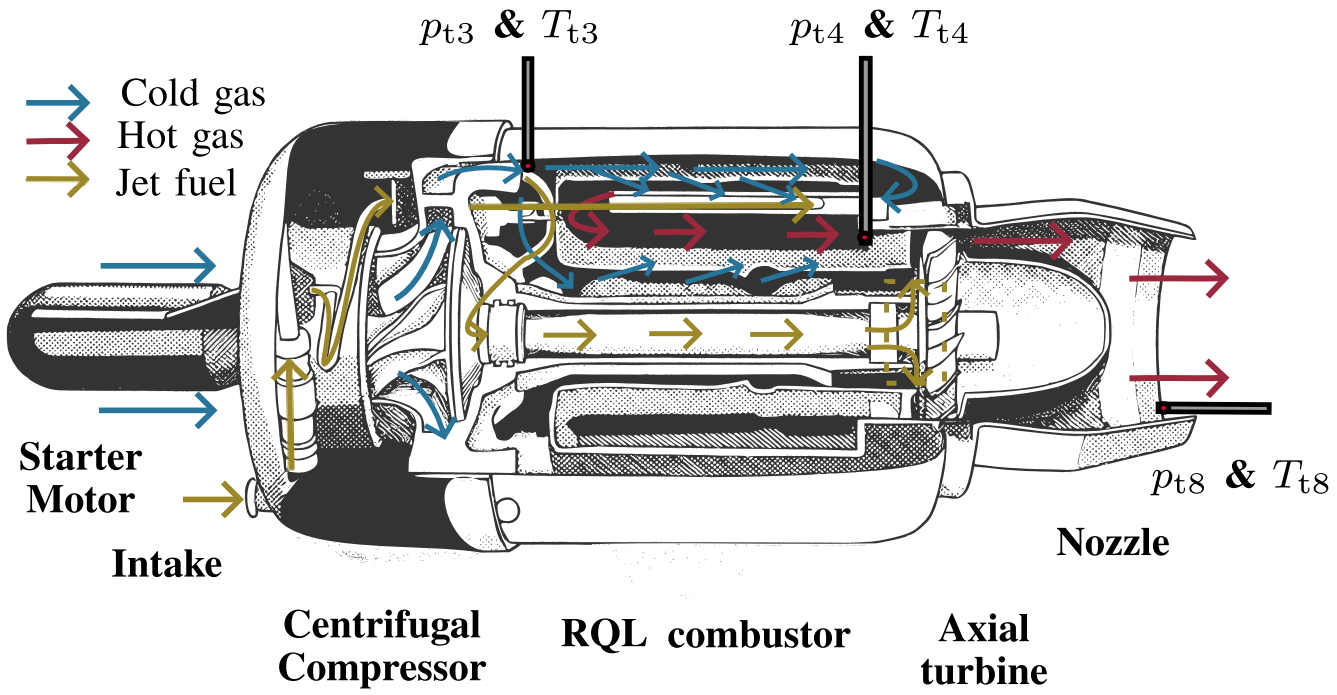
How can a laboratory-scale micro-turbojet test rig, configured for back-to-back testing of conventional and "drop-in" alternative jet fuels, be successfully commissioned to generate useful experimental data?





Methodology: The Micro-Turbojet Engine

JetCat P200-SX Micro-Turbojet:



Engine Specifications:

Max RPM	112,000
Max Thrust (N)	230
Pressure Ratio	4
Mass Flow (kg/s)	0.45
Fuel Flow @ Max RPM (kg/s)	0.0096
Nozzle Exit Area (m^2)	0.0028

- The engine utilizes **open-loop** lubrication system [10].

[10] Lieder et al., *ASME Turbo Expo*, 2026

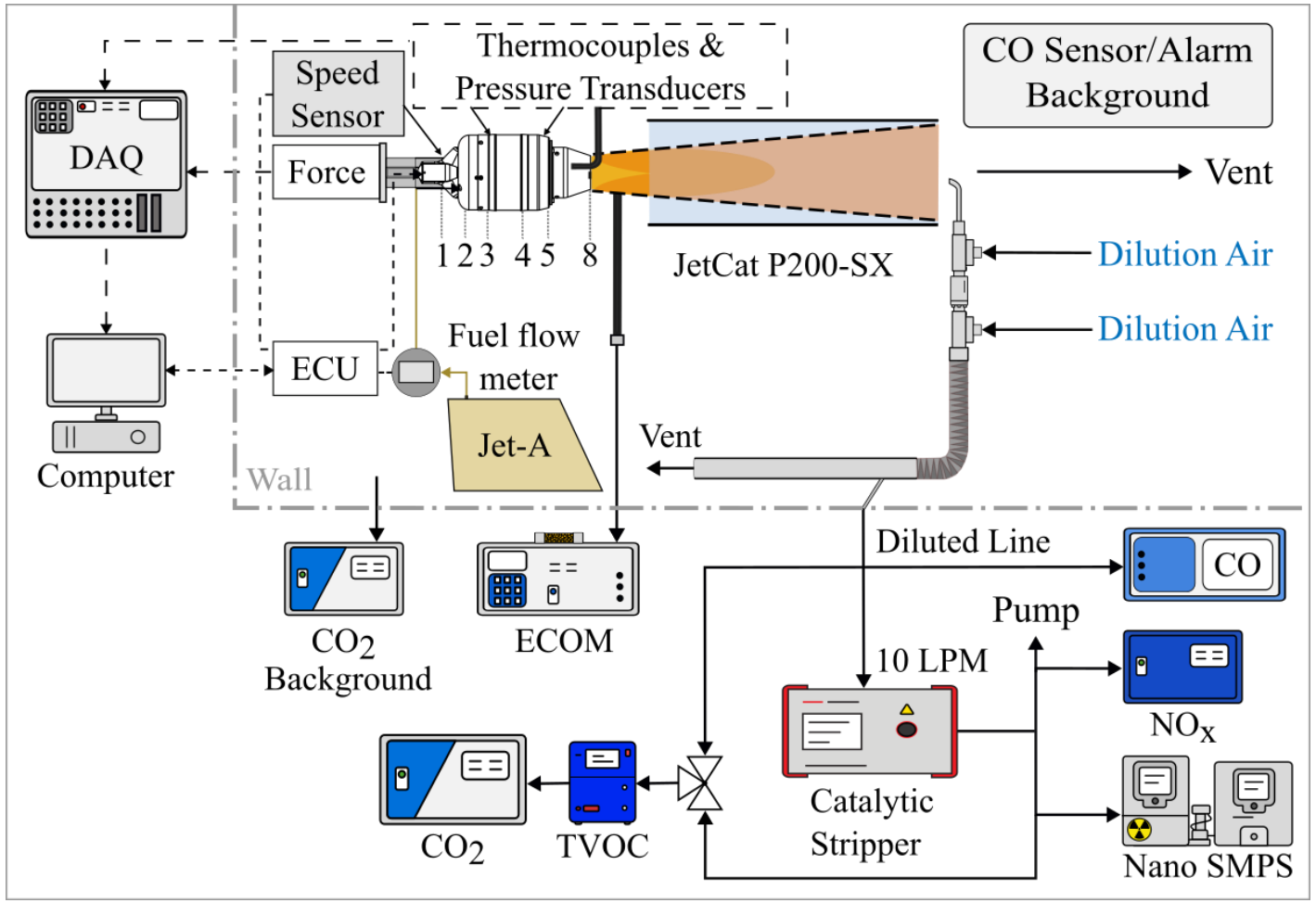
Methodology: Experimental Setup

- **Gas-Phase Measurement**
  - Dedicated NOx, CO, and CO_2 analyzers on the diluted line.

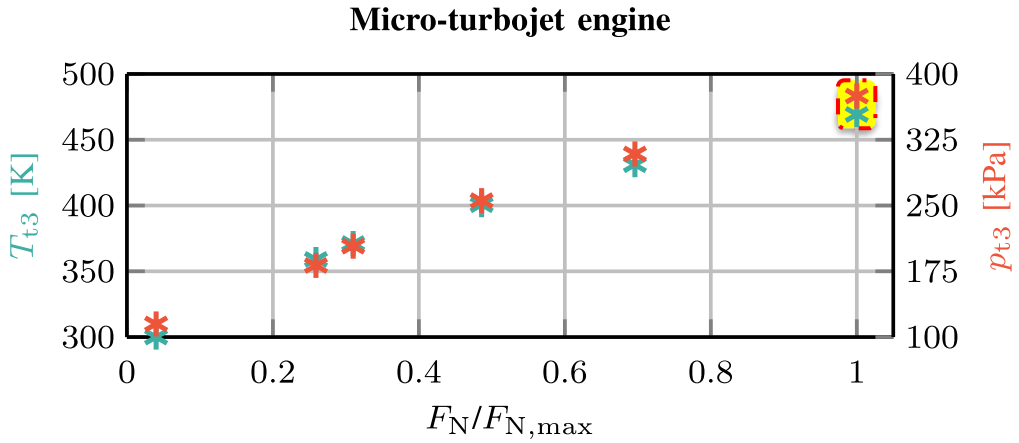
$$EI_i\ [\mathrm{g/kg_{fuel}}] = \frac{1000 \cdot M_i \cdot \eta_{\mathrm{comb}} \cdot Y_C}{M_C} \cdot \frac{x_i^{DL}}{\left(x_{\mathrm{CO_2}}^{DL} - x_{\mathrm{CO_2}}^{bg}\right) + x_{\mathrm{CO}}^{DL}}$$
$$\textbf{with } i = \mathrm{CO_2},\ \mathrm{CO},\ \mathrm{NO}_x$$

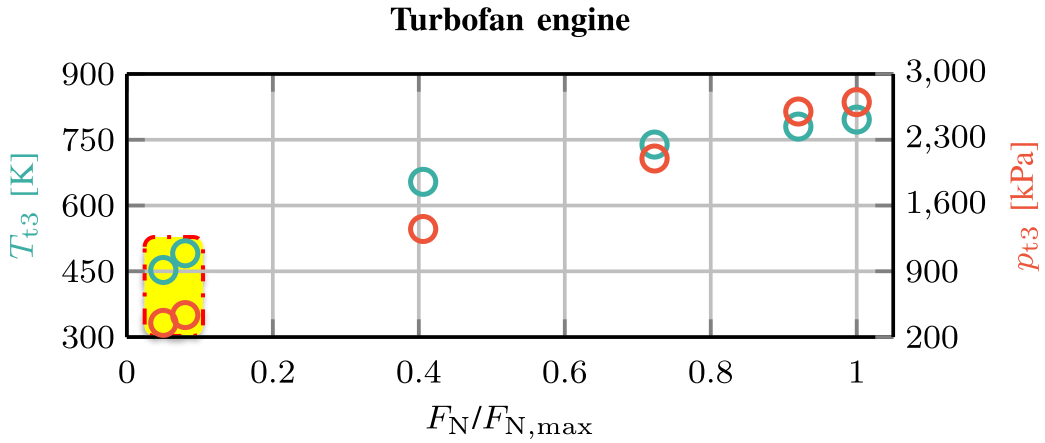
- **Particle Phase Measurement**
  - Measures via SMPS (nano-DMA) downstream of a **Catalytic Stripper**.

$$\frac{dEI_{n,vPM,N}}{d\log_{10} d_p}\ [\#/\mathrm{kg_{fuel}}] = \frac{10^{12} \cdot \frac{dN_{v,\mathrm{corr}}}{d\log_{10} d_p} \cdot \eta_{\mathrm{comb}} \cdot Y_C \cdot R \cdot T_{\mathrm{ref}}}{\left[\left(x_{\mathrm{CO_2}}^{DL} - x_{\mathrm{CO_2}}^{bg}\right) + x_{\mathrm{CO}}^{DL}\right] \cdot M_C \cdot p_{\mathrm{ref}}}$$



RESULTS: MICRO-TURBOJET VS TURBOFAN





Direct operating-point comparison between the JetCat and a full-scale V2500 turbofan is constrained by scale.

However, a specific thermodynamic intersection exists:

"Micro-Max / Macro-Idle": This thermodynamic intersection establishes a physically valid anchor point for the performance.

Very similar combustor inlet conditions!

Combustor inlet conditions

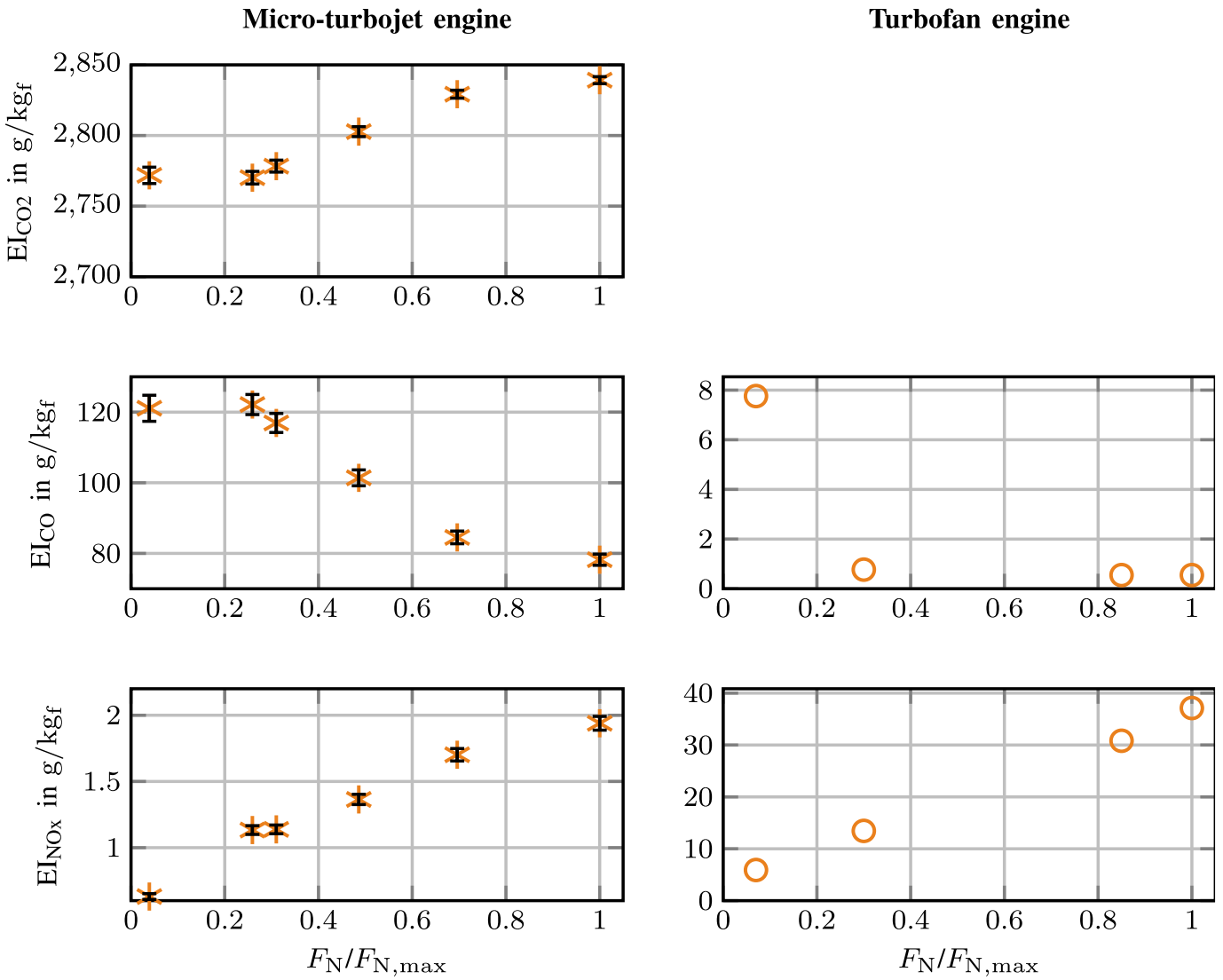
	Micro-turbojet (Max Thrust)	Full-scale turbofan (Idle)
T_{t3}	469 K	453 K
p_{t3}	374.8 kPa	351 kPa
OPR	~4.0	~3.5

Intensive properties (T, p, ϕ) are very similar (scale-independent).

RESULTS: GASEOUS EI

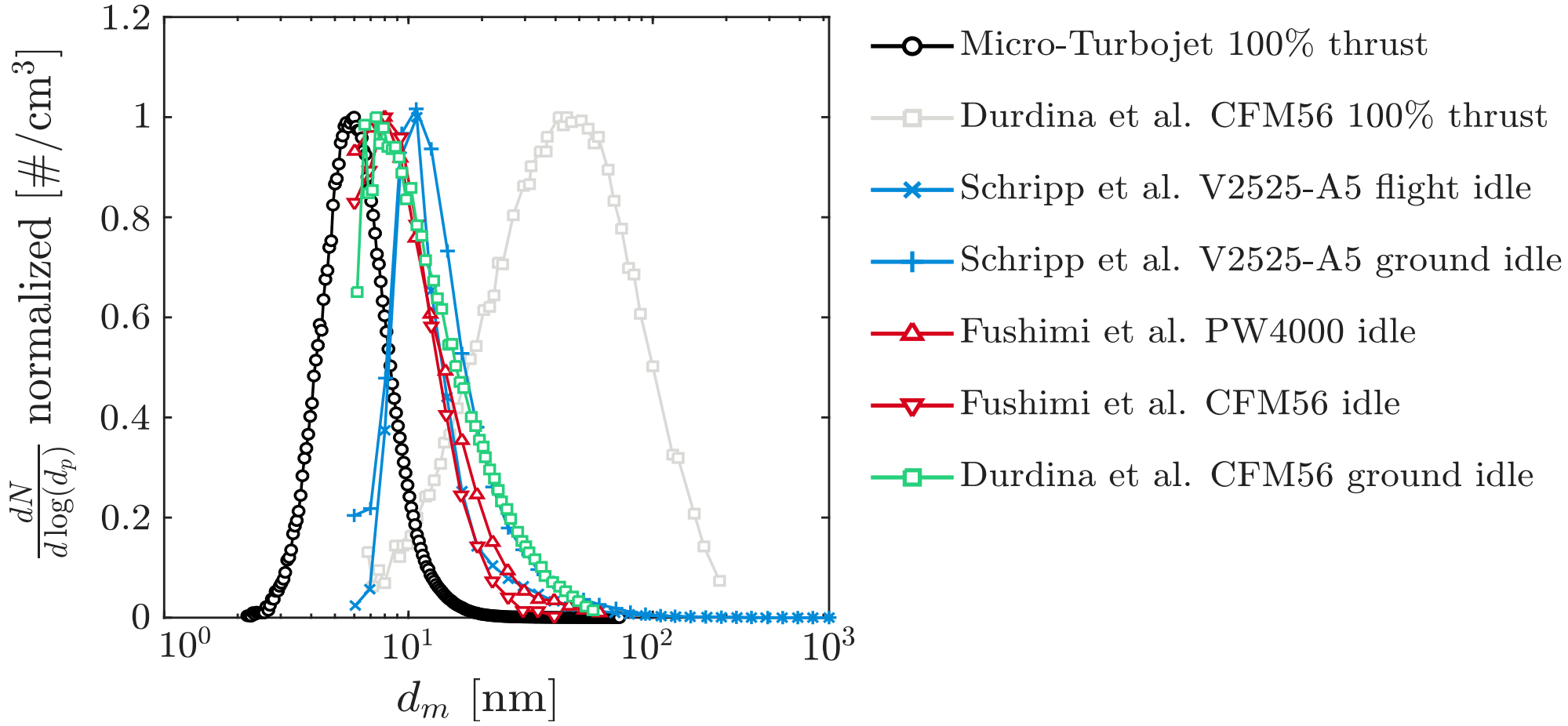
- ICAO does not provide CO_2 reference data.
- **CO decreases** with increasing thrust, while **NO_x increasing** with thrust

Qualitatively: Preservation of directional trends.



RESULTS: NON-VOLATILE PM

Comparison of loss corrected normalized particle size distributions:



Summary & Future work

Summary

Generate Reference Data with Jet A-1
Performance, & Emissions across operating points.

Capture Key Trends
CO_2, CO, NO_x, nvPM vs. operating point.

Micro-Max/Macro-Idle
Identified thermodynamic intersection!

Compare Emissions
- Qualitative trends preserved in gaseous emissions.
- Micro-turbojet nvPM PSD consistent with idling turbofan engines

Demonstrate Test-rig Benefits
Assessment of micro-turbojet test rig & thermodynamic regime matching.

Future Work

Update Experimental Setup

More Fuels, FT-SPK, SAF's

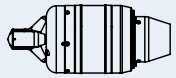
Employ a Coupled framework with CRN & Sectional Soot model

Additional Instrumentation: FTIR, AMS, TEM, ELPI



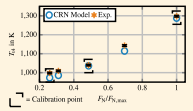
Scaling studies, extrapolate to full-scale engine emissions

Coupled Framework



Micro-Turbojet Experiment Emissions Data

+



CRN Model Emissions Data
(CO_2, CO, NO_x, nvPM) formation

Model Assessment
Trends + gaps
Quantify Uncertainties

- **No Regulatory Bottlenecks**
- **Very modest fuel requirement (tens of liters)**
- **Cost-effective data generation**
- **Applicable for various fuel types, and large emission assessments**

Acknowledgments

- UBC Clean Energy Research Center (CERC)
- The NSERC Discovery Grant Program
- The Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy
- SE²A - Sustainable and Energy-Efficient Aviation
- KAUTE Foundation through the Researchers Abroad program and the GIANT project funded by Business Finland, Helsinki Region Environmental Agency HSY

Supervisor: Prof. Steven Rogak

Individuals: Prof. Patrick Kirchen, Dr.-Ing. Jan Göing, Daniel Leider (PhD Candidate TUBS), Henna Lintusaari (PhD Candidate Tampere University), Nishan Sapkota (PhD Candidate UBC), and everyone at CERC for letting us operate a jet engine!

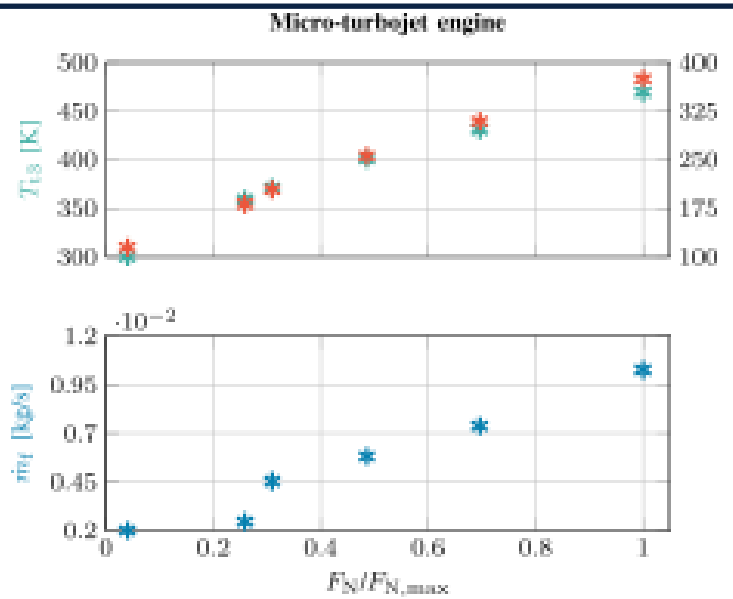


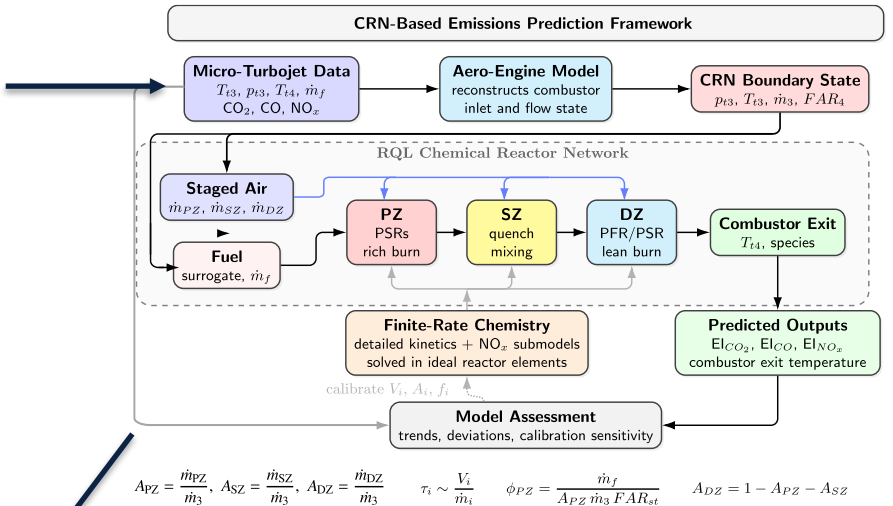
Thank you, Questions?

APPENDIX

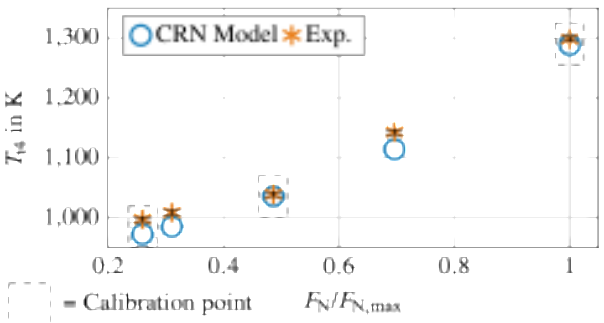
APPENDIX: JET A-1 PERFORMANCE DATA

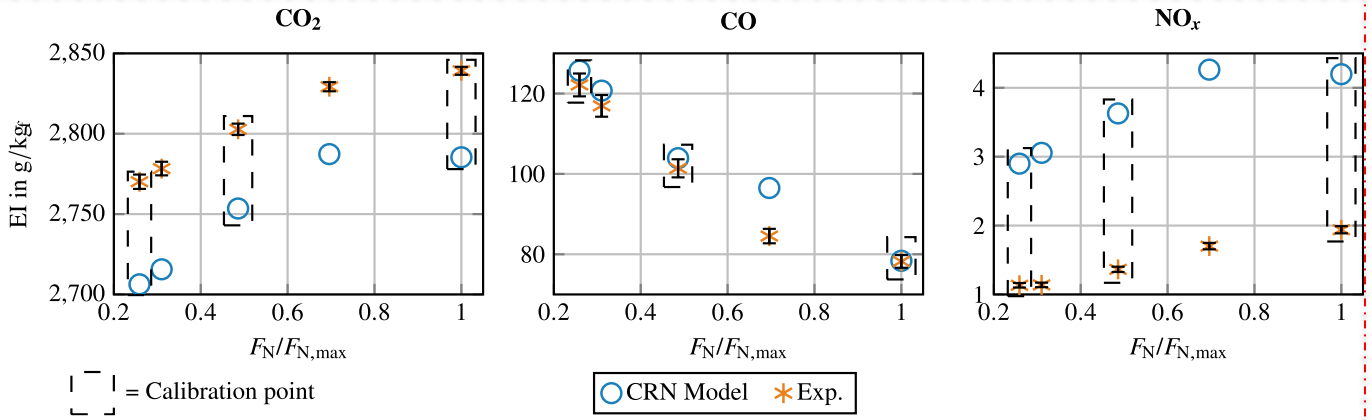
Results from complementary work:
Here Tt_3, pt_3, $\dot{m}_f$, are used to calibrate the aero-engine performance model, where it reconstructs the combustor inlet state, which are used as the boundary conditions for the CRN model.





Results (Lieder et al., *ASME Turbo Expo*, 2026)





Appendix: Physics Based Model

Reduced-order physics-based framework

- Captures fuel and operating-point-dependent emissions sensitivities.
- Altitude effects are represented physically.
- Calibrated using experimental engine data.

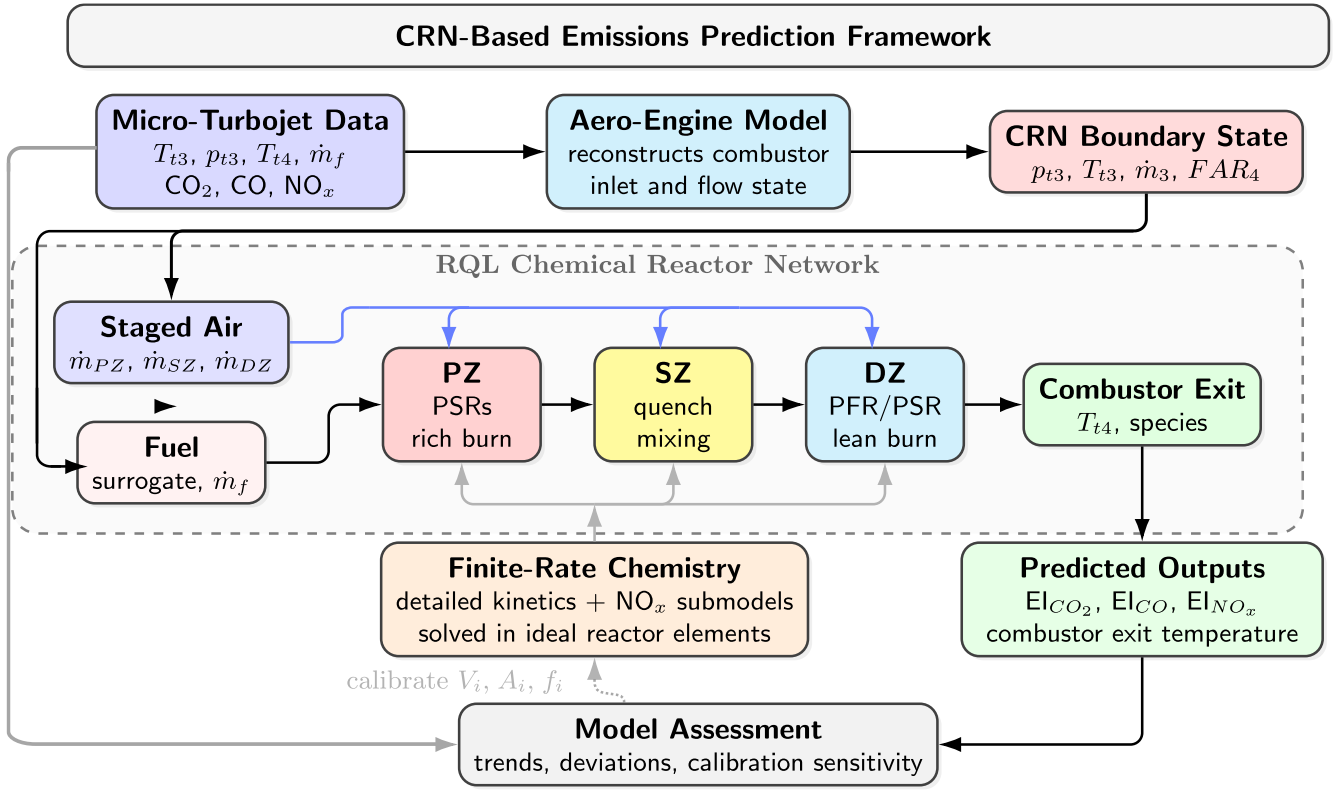
Aero model provides combustor inlet state:

- Pressure, temperature, mass flow rates, & fuel to air ratio.

RQL combustor represented using ideal reactors:

- Finite-rate chemistry predicts CO_2, CO, NO_x, nvPM

Enables fuel-and operating-point-sensitive emissions prediction & feasible for large emissions assessments



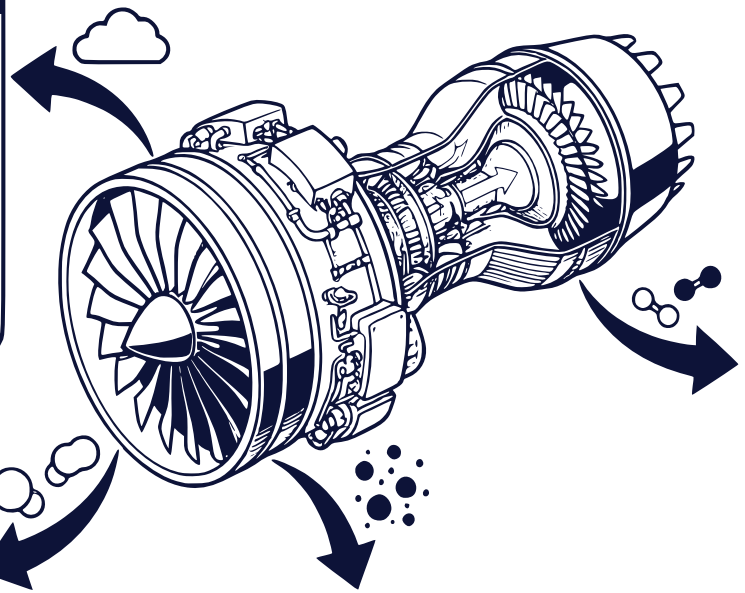
$$A_{PZ} = \frac{\dot{m}_{PZ}}{\dot{m}_3},\ A_{SZ} = \frac{\dot{m}_{SZ}}{\dot{m}_3},\ A_{DZ} = \frac{\dot{m}_{DZ}}{\dot{m}_3} \qquad \tau_i \sim \frac{V_i}{\dot{m}_i} \qquad \phi_{PZ} = \frac{\dot{m}_f}{A_{PZ}\,\dot{m}_3\,FAR_{st}} \qquad A_{DZ} = 1 - A_{PZ} - A_{SZ}$$

Appendix: Aero Engine Emissions

Products of Complete Combustion (Greenhouse Gases):



Carbon dioxide (CO_2) & Water vapor (H_2O): are the primary natural byproducts of hydrocarbon combustion. Both are GHGs. H_2O plays a critical role in contrail formation at high altitudes.



Products of Incomplete Combustion (Lower Power):



Carbon Monoxide (CO) & Unburned Hydrocarbons (UHC): Greatest concentrations predominantly generated during low-power operation (e.g., idle), where combustion efficiency is lowest.

High-Temperature Byproducts (Higher Power):



Oxides of Nitrogen (NOx), (NO & NO_2): formed via the thermal reaction of atmospheric **N_2** and **O_2** at extremely high temps (**>1800 K, Thermal (Zel'dovich) mechanism**), during high-power settings like take-off and climb.

Particulate Matter (PM) & Impurities:



Non-volatile PM (nvPM): Ultrafine black carbon (soot) particles formed in fuel-rich regions of the combustor.
Volatile PM (vPM): Formed when gaseous precursors (like volatile organic compounds) cool and condense after exiting the engine.

Aero engine emissions (depend on flight phase, & fuel composition)

Appendix: Aero Emissions Regulations

How do international bodies regulate these emissions and the fuels that produce them?

The Regulatory Landscape:

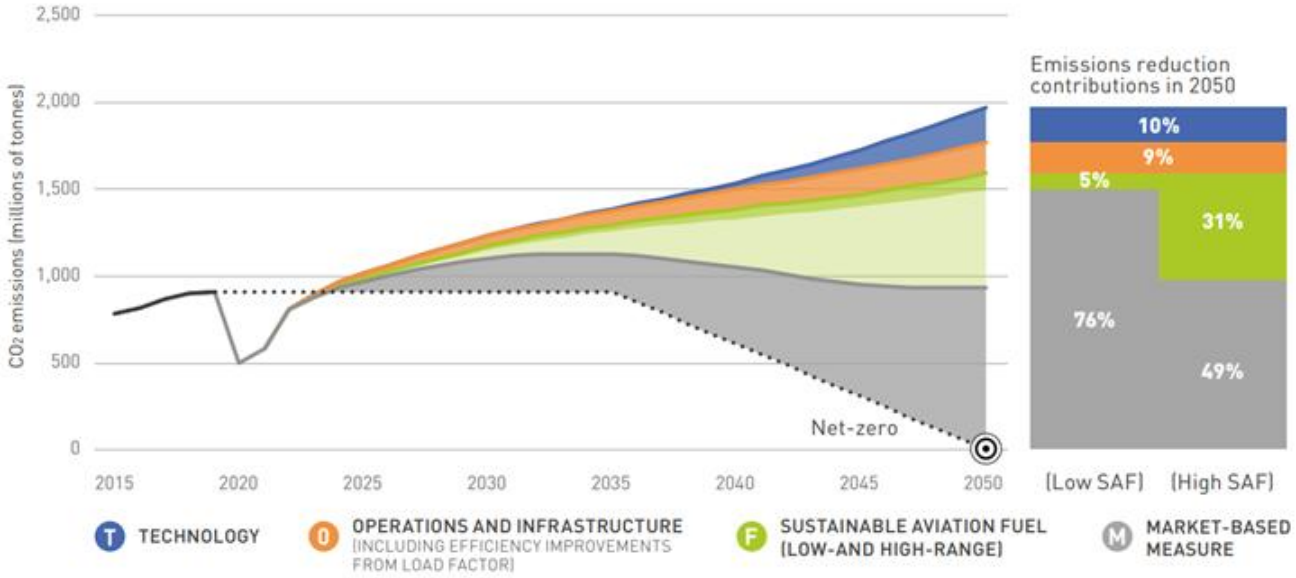
- **Engine Certification (ICAO Annex 16):** Standards for NO_x, nvPM, CO, and UHC are set by the CAEP (Committee on Aviation Environmental Protection) and enforced by national authorities like the Federal Aviation Administration (FAA) or Transport Canada Civil Aviation (TCCA) [8].

- **Fuel Certification (ASTM International): *ASTM D1655:** The standard for conventional Jet A/A-1 [13].
    - **ASTM D7566:** The critical specification for Sustainable Aviation Fuels (SAF), certifying them as "drop-in" ready [14].

[9] (ICAO Annex 16), [13] (ASTM D7566-25a), [14] (ASTM D1655-25a)

APPENDIX: SUSTAINABLE AVIATION FUELS

The Near-Term Solution: Sustainable aviation fuels (SAFs) are the primary viable pathway to decarbonize aviation, offering "drop-in" replacements sourced from biogenic or waste feedstocks [8, 35].



Industry targets to achieve “net zero” CO2 emission. [16].

- **Regulatory Hurdles:** Aviation operates under strict safety margins. Aircraft cannot legally fly on a new fuel unless it has rigorous ASTM International certification (e.g., ASTM D7566) [13].
- **The Reality:** Many promising, highly sustainable SAF pathways are not yet certified for commercial use, meaning they cannot be deployed [15].

[8] (ICAO Annex 16), [13] (ASTM D7566-25a), [15] (ICAO 2017), [16] (ATAG 2021)

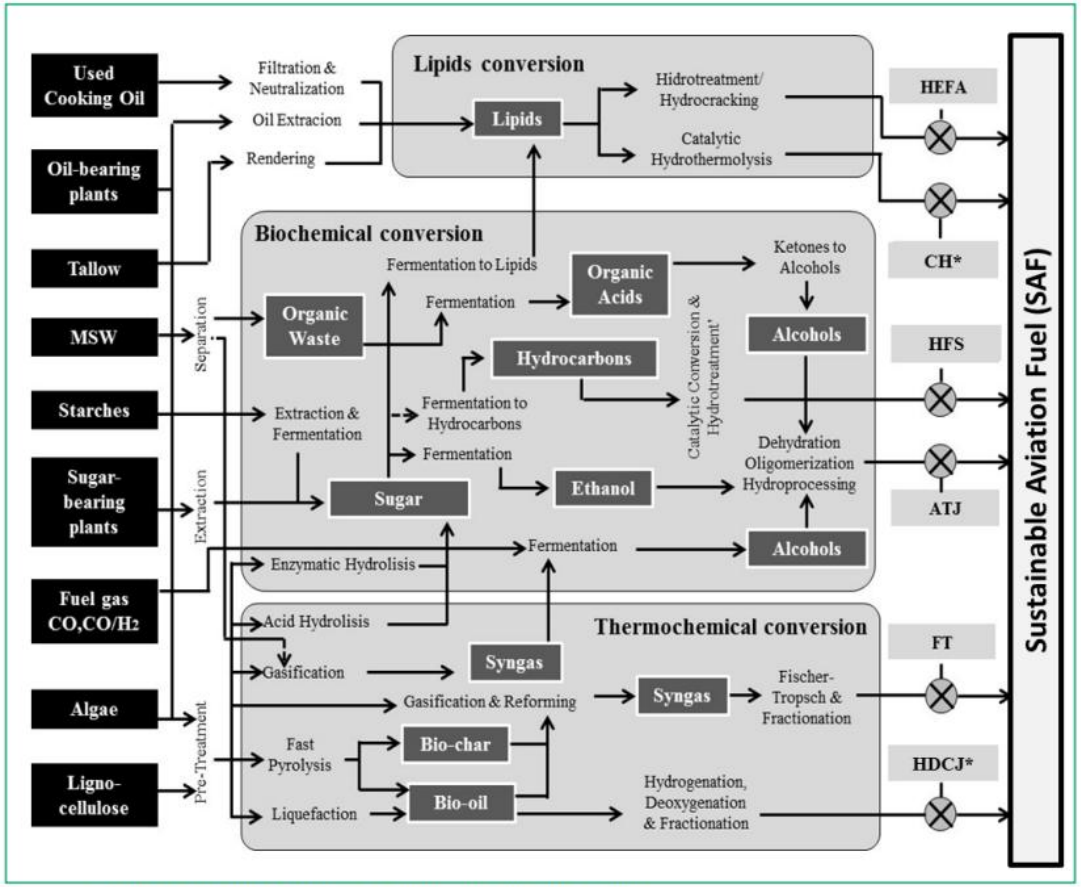
Appendix: Sustainable Aviation Fuels

"Drop-In": While marketed as "drop-in" replacements, SAFs are different from conventional Jet Fuel (Jet A-1) & one another. Their chemistry is dependent on the feedstock and processing pathway.

Key Chemical Differences:

- **SAFs differ in Aromatic Content**
- **SAFs differ in Hydrogen-to-Carbon (H/C) Ratio**
- **SAFs differ in Volatility**

The Impact: The chemical composition dictate the emissions. Because SAFs are all different, we cannot assume any two SAFs will behave the same way in an engine.



*CH: Catalytic hydrothermolysis – not yet approved
HDCJ: Hydrotreated depolymerized cellulosic to jet – not yet approved

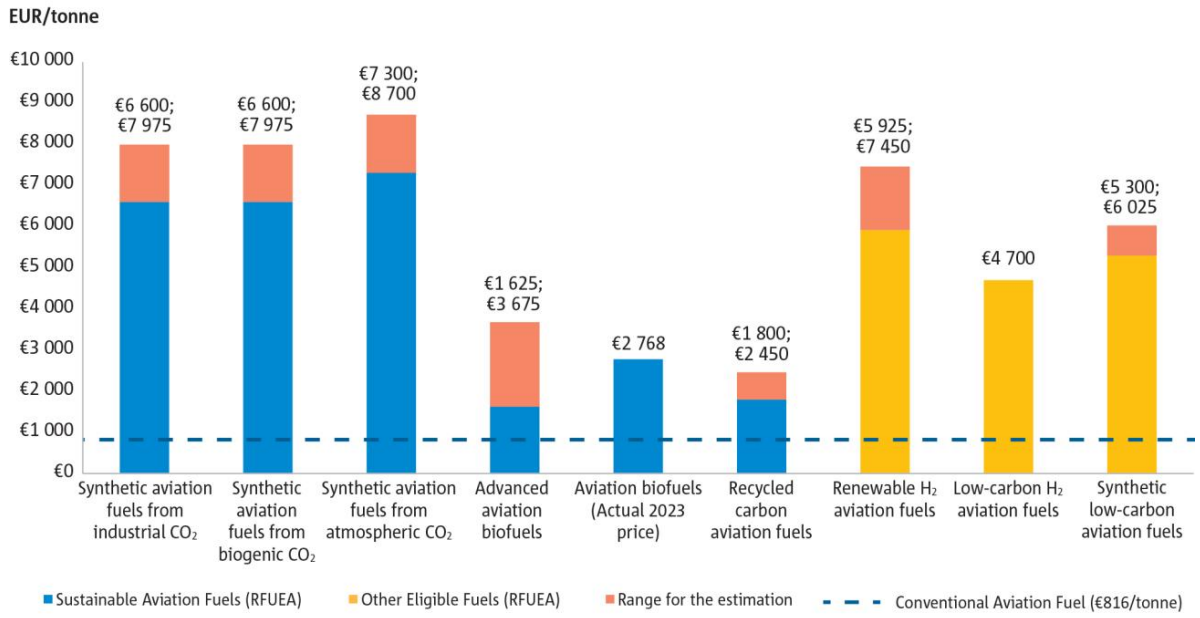
ICAO – Sustainable Aviation Fuels Guide: Different processing routes can be adopted to convert feedstocks into a liquid drop-in SAF. Several combinations of feedstocks and conversion processes have been proposed. Only some of them have been certified to be used as a drop-in fuel. [15].

[8] (ICAO Annex 16), [13] (ASTM D7566-25a), [15] (ICAO 2017)

Appendix: Sustainable Aviation Fuels

To characterize the emissions from fuels we need extensive data on how they burn. Gathering this emissions data using full-size commercial turbofans is a major bottleneck:

Regulatory constraints: Strict safety protocols make testing experimental, uncertified fuels on large commercial hardware incredibly restrictive.



Estimated prices and production costs in 2023 for ReFuelEU Aviation eligible fuels [17].

Fuel Intensive: Requires tens of thousands of liters of experimental fuel, which is nearly impossible to produce for early-stage novel SAFs.

Prohibitively Expensive: These novel SAFs are orders of magnitude more expensive than Jet A-1 testing costs tens of thousands to hundreds of thousands of dollars per campaign for simply fuel costs, not accounting for the requisite hardware to burn and characterize these fuels.

[17] (EASA 2022)

Appendix: Sustainable Aviation Fuels

"Drop-In": While marketed as "drop-in" replacements, SAFs are fundamentally different from one another. Depending on the feedstock and pathway, some are excellent, while others have incredibly poor combustion characteristics.

Key Chemical Differences:

Aromatic Content: Often much lower in SAFs. This reduces soot/nvPM formation but causes critical issues with engine seals shrinking and leaking.

Hydrogen-to-Carbon (H/C) Ratio: Typically higher, which fundamentally alters the flame temperature and structure in the combustor.

Lower Heating Value (LHV): Differences in energy density directly impact fuel flow rates and overall engine performance.

The Impact: These chemical shifts dictate the emissions. Because SAFs are all different, we cannot assume any two SAFs will behave the same way in an engine.

[15] (ICAO 2017)

Appendix: Back-to-back Testing

Back-to-back testing: Consecutively testing different fuels under identical ambient and operating conditions to strictly isolate fuel-property effects from engine effects.

Why?

Isolates the Variables: Testing consecutively minimizes the impact of shifting environmental factors (temperature, humidity, atmospheric pressure) on combustion.

Proves "Fuel is the Factor": Ensures any observed shifts in emissions or performance are driven purely by fuel chemistry (e.g., aromatics, H/C ratio) and not by changes in engine operating conditions.

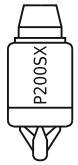
Captures Relative Sensitivities: Focuses on generating highly accurate differential data (e.g., EI_{CO} between SAF and Jet A-1) rather than just absolute magnitudes.

Enables Predictive Modeling: Provides the tightly controlled, physically grounded datasets required to accurately calibrate numerical scaling tools, such as Chemical Reactor Network (CRN) models.

APPENDIX: CALCULATION OF EMISSION INDICES

Instrument	Species	Units	Accuracy
2B 405 nm	NO_x	PPB	±2 %
T300U	CO	PPM	±1 %
LI-850	CO_2	PPM	±1.5 %

A. EXPERIMENTAL DATA: CARBON-BALANCE METHOD



Inputs (measured at diluted line)

$x^{DL}_{CO_2}, x^{DL}_{CO}, x^{DL}_{NO_x}$ (dry mole fractions)

$\frac{dN_{nv,corr}}{d\log_{10} d_p}$ (corrected nvPM number size distribution)

$x^{bg}_{CO_2}$ (background CO_2)

1 Gaseous species EI

$$EI_i\ [\text{g/kg}_{\text{fuel}}] = \frac{1000 M_i\, \eta_{comb} Y_C}{M_C} \frac{x_i^{DL}}{(x^{DL}_{CO_2} - x^{bg}_{CO_2}) + x^{DL}_{CO}}$$

with $i = CO_2, CO, NO_x$

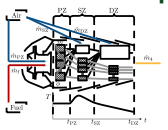
2 nvPM number EI (size-resolved)

$$\frac{dEI_{nvPM,N}}{d\log_{10} d_p}\ [\#/\text{kg}_{\text{fuel}}] = \frac{10^{12}\ \frac{dN_{nv,corr}}{d\log_{10} d_p}\ \eta_{comb} Y_C R T_{ref}}{\left[(x^{DL}_{CO_2} - x^{bg}_{CO_2}) + x^{DL}_{CO}\right] M_C p_{ref}}$$

$dN_{nv,corr}/d\log_{10} d_p$: nvPM number size distribution corrected for dilution, thermophoretic and size-dependent losses [#/cm^3]

B. CRN MODEL: CRN EI CALCULATION METHOD

Inputs (from aero-engine model & CRN)



$\dot{m}_{i,out}, Y_{i,out}$ (gas mass fractions),

$\frac{d\dot{N}_{soot}}{d\log_{10} d_p}$ (soot number size distribution)

$\dot{m}_f$ (fuel mass flow rate)

1 Gaseous species EI

$$EI_i\ [\text{g/kg}_{\text{fuel}}] = 1000 \frac{\dot{m}_{i,out}}{\dot{m}_f}$$

$$\dot{m}_{i,out} = Y_{i,out} \dot{m}_{out}$$
with $i = CO_2, CO, NO_x$

2 nvPM number EI (size-resolved)

$$\frac{dEI_{nvPM,N}}{d\log_{10} d_p}\ [\#/\text{kg}_{\text{fuel}}] = \frac{1}{\dot{m}_f} \frac{d\dot{N}_{soot}}{d\log_{10} d_p}$$

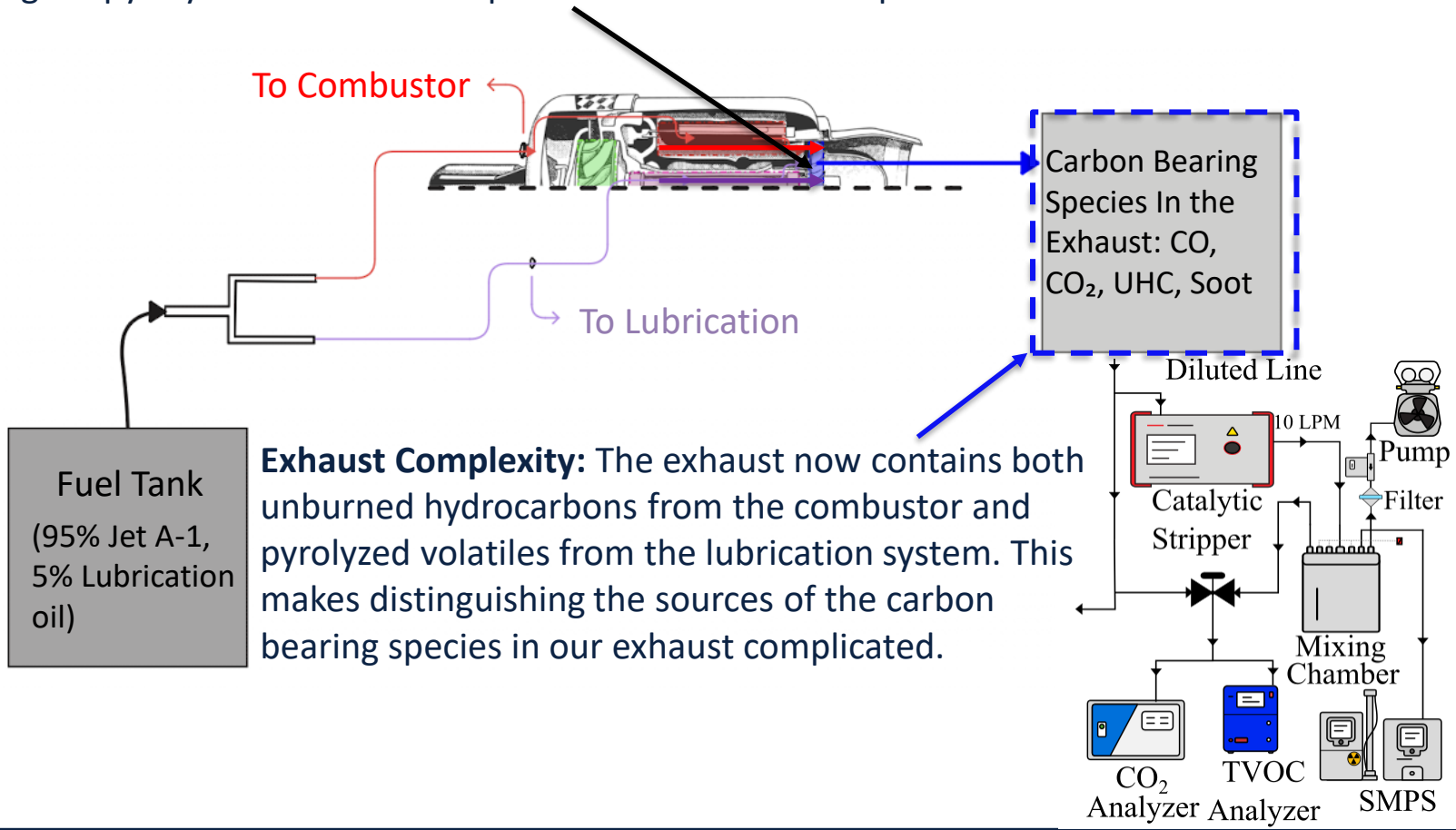


CRN soot model (sectional approach):

Finite-rate chemistry coupled with a discrete sectional soot model (21 size bins; $d_p \approx 2.02$ nm to 202.12 nm) predicts soot number and mass across particle sizes.

APPENDIX: INDIRECT UHC MEASUREMENT

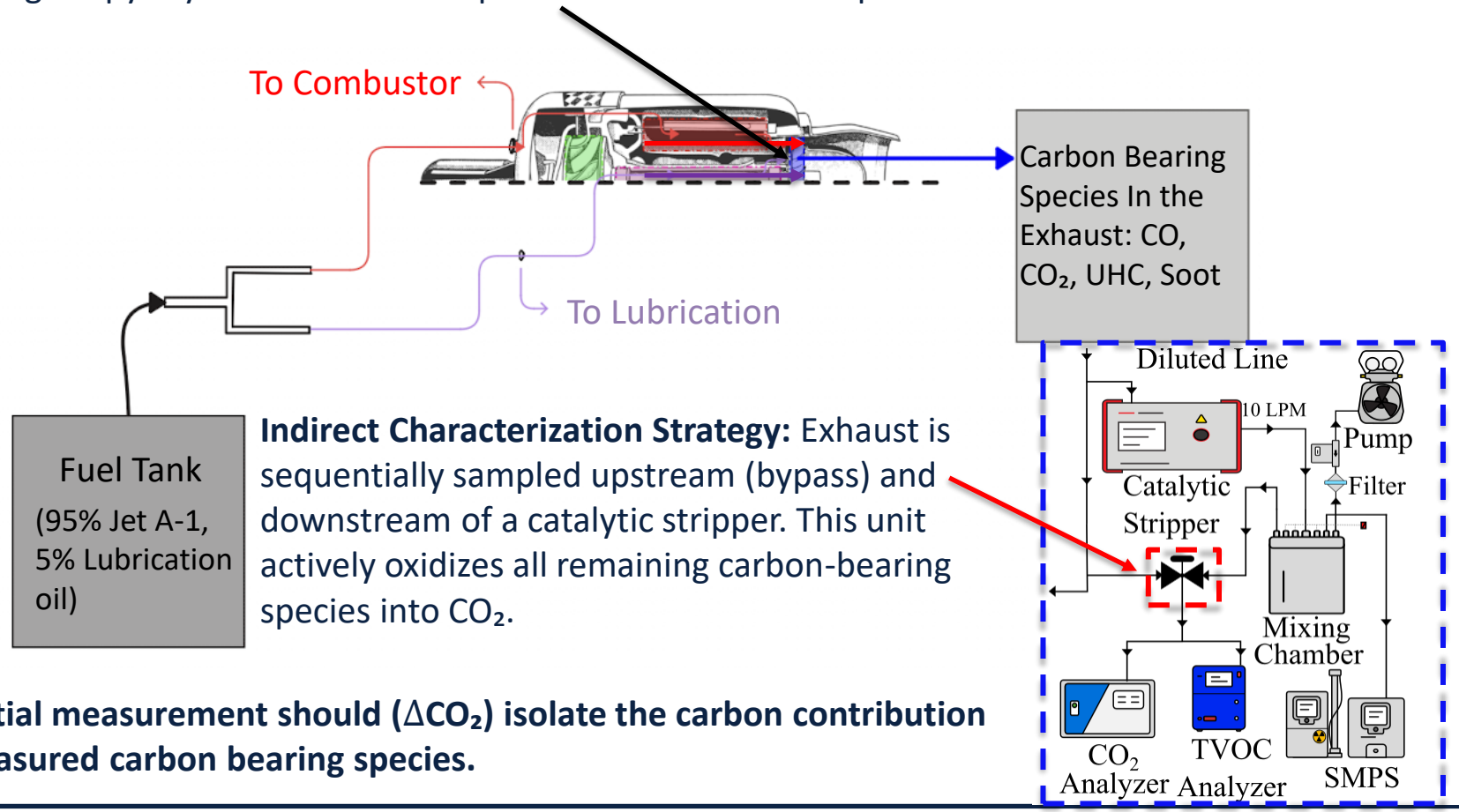
Open-loop: The system splits a 95% Jet A-1 / 5% lubrication oil blend between the combustor and the open-loop lubrication path. Lubricant is ejected directly into the gas path upstream of the axial turbine ($T_{t4} \approx 1200\ K\ -\ 1400\ K$). Within this high-temperature, oxygen-depleted environment, the lubrication oil likely undergoes pyrolytic thermal decomposition rather than complete oxidation.



Exhaust Complexity: The exhaust now contains both unburned hydrocarbons from the combustor and pyrolyzed volatiles from the lubrication system. This makes distinguishing the sources of the carbon bearing species in our exhaust complicated.

APPENDIX: INDIRECT UHC MEASUREMENT

Open-loop: The system splits a 95% Jet A-1 / 5% lubrication oil blend between the combustor and the open-loop lubrication path. Lubricant is ejected directly into the gas path upstream of the axial turbine ($T_{t4} \approx 1200\ K\ -\ 1400\ K$). Within this high-temperature, oxygen-depleted environment, the lubrication oil likely undergoes pyrolytic thermal decomposition rather than complete oxidation.



Indirect Characterization Strategy: Exhaust is sequentially sampled upstream (bypass) and downstream of a catalytic stripper. This unit actively oxidizes all remaining carbon-bearing species into CO_2.

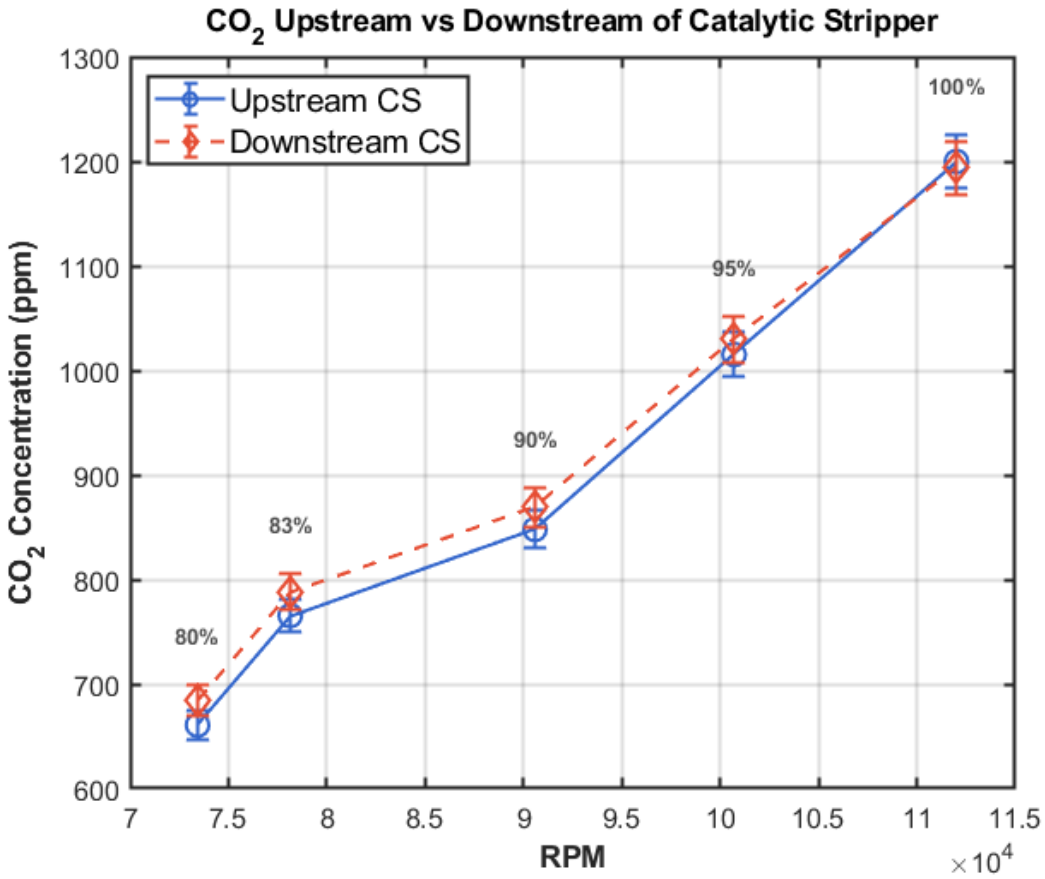
The differential measurement should (ΔCO_2) isolate the carbon contribution of the unmeasured carbon bearing species.

APPENDIX: INDIRECT UHC MEASUREMENT

- **The ΔCO_2 Signal:** Downstream CO_2 (red) is higher than upstream (blue) but decreasing with load, confirming the oxidation of unburned carbon species.

- **Load Dependency:** The measurable difference (ΔCO_2) is small and decreases as engine RPM increases.

It is interesting to note, that while the ΔCO_2 decreases with increasing RPM, the amount of lubrication oil flow rate also increases with the engine RPM.



Functional validation of the indirect UHC measurement system. Diluted exhaust gas is sequentially routed through a bypass line and a heated catalytic stripper using a 2-way switching valve. The stripper actively forces the complete oxidation of any surviving unburned fuel and pyrolyzed lubrication volatiles into measurable CO_2. This differential methodology isolates the total volatile carbon contribution without requiring complex chemical speciation.

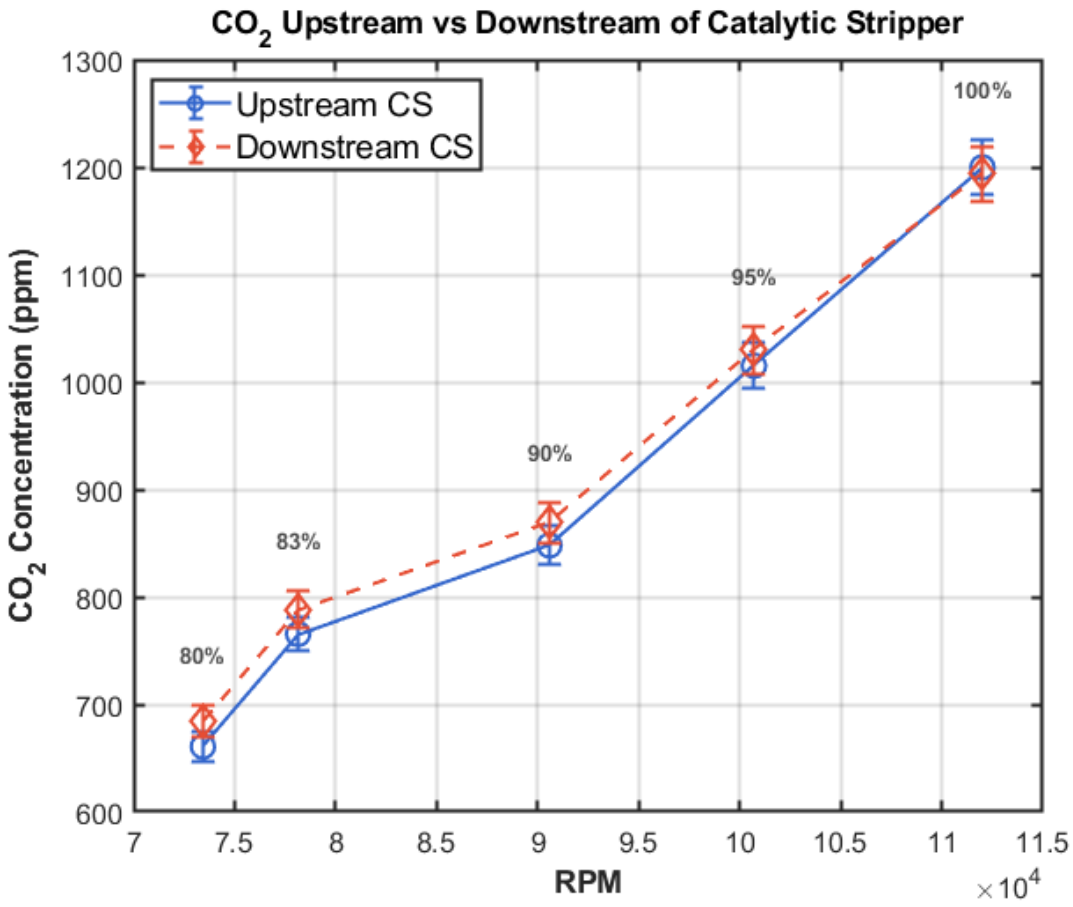
APPENDIX: INDIRECT UHC MEASUREMENT

System Split: Consistently, approx. 88% of the total flow reaches the primary combustor, while approx. 12% feeds the open-loop lubrication path.

At 100% thrust, the absolute mass of oil ejected into the exhaust is roughly four times higher than at idle (90 mL/min vs. 22 mL/min).

Thermal Oxidation: Despite this massive increase in injected oil, the ΔCO_2 gap shrinks to almost zero at max thrust.

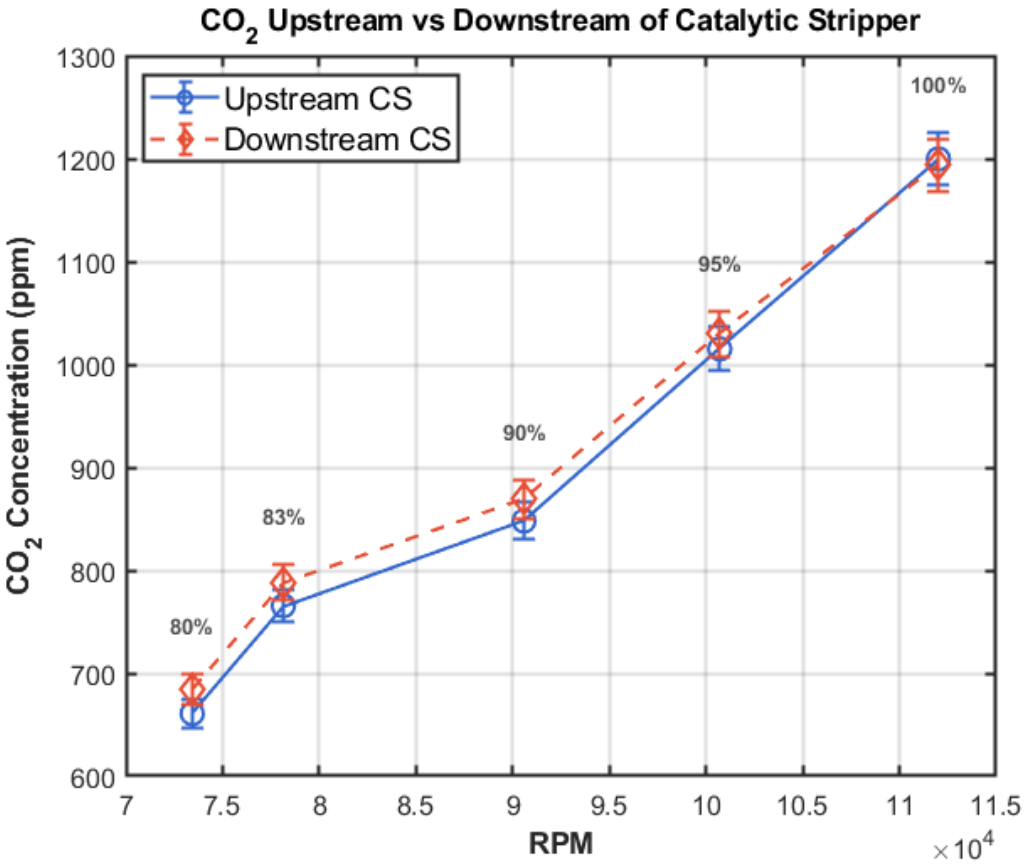
Physical Conclusion: At maximum load, extreme turbine inlet temperatures $T_{t4} \approx 1200\ K - 1400\ K$) likely cause the slipstream mixture to auto-ignite and fully oxidize before reaching the exhaust plane.



APPENDIX: INDIRECT UHC MEASUREMENT

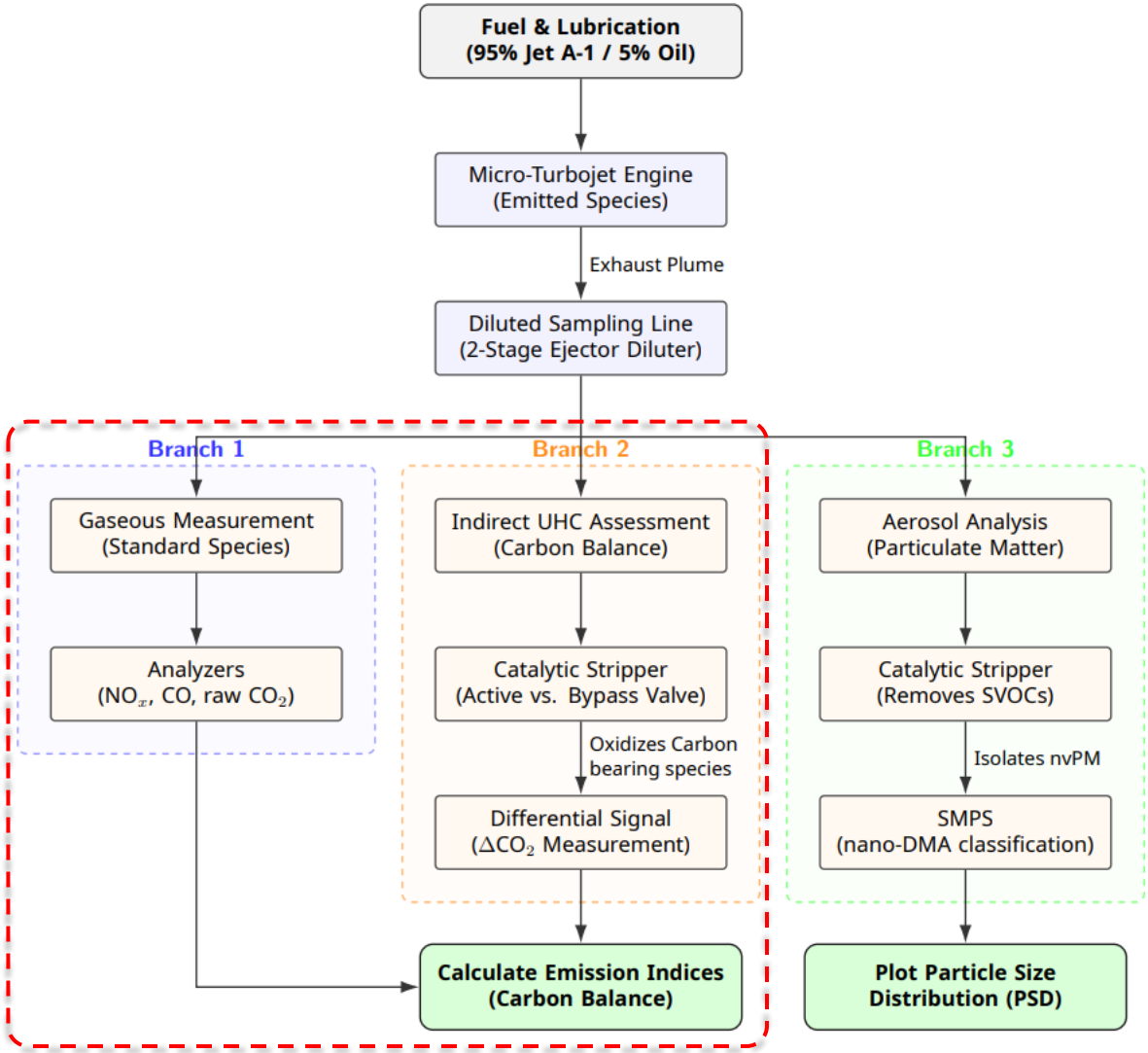
Low-Load Inefficiency is the Real Culprit: At lower RPMs (like 33k to 42k), temperatures are lower. The 22 mL/min of lubrication oil doesn't fully burn, and the primary combustor is operating inefficiently, dumping out primary UHCs. That's is likely why the ΔCO_2 gap is widest there.

If 12% of the fuel was just slipping out of the engine unburned, the carbon balance would be off by 12%, and we had a max deviation of 6% at the lowest operating point.

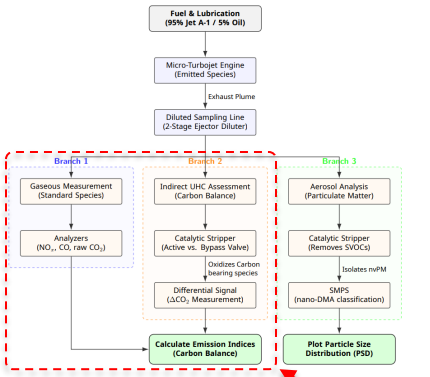


Because the catalytic stripper proves the unmeasured carbon is negligible at high loads, we have proof that treating the exhaust as "fully oxidized" at the micro-max condition is a perfectly valid assumption.

APPENDIX: DATA COLLECTION



APPENDIX: DATA PROCESSING



Raw data processing pipeline for Branch 1 & 2

Phase 1: Signal Pre-Processing & Temporal Alignment

1. Raw Data Ingestion
(Perf, ECOM, LICOR, NO_x/CO Analyzers)

2. Time Alignment & Interpolation
(Correct for sampling time lags & Align to 1Hz Grid)

3. Engine Ignition Trimming
(Detect 1st Peak >30k RPM, Apply 5s Buffer, Set $t = 0$)

4. Steady-State Identification
(Rolling 30s Window, Filter by σ Threshold & ROC)

Phase 2: Categorization of Operating Points

Branch 1: Gaseous Mapping
(Extract NO_x, CO, and baseline CO_2 from steady regions)

Branch 2: UHC Mapping
(Isolate ΔCO_2 variance: Upstream vs. Downstream Stripper)

Phase 3: Final Analytics & Uncertainty

5. Calculate Emission Indices (EI)
(Carbon Balance)

6. Calculate Uncertainty (95% CI)
(Propagating Instrument Uncertainties)

7. Approach Uncertainty (Planned)
(Comparing operating point data between runs to find the uncertainty of the approach itself)

Appendix: Physics surrounding comparison

Micro (JetCat)

- **Surface-to-volume:** High (significant wall quenching).
- Thus, **lower peak flame temperature**: Thermal NOx is exponentially dependent on temperature. We see **much lower EI NOx** for micro-turbojet which is consistent.
- Residence Time: ≈ 2 ms, this will limit CO oxidation.
- Max overall pressure ratio (OPR): 4

Macro (V2500)

- **Surface-to-Volume:** Low (Adiabatic core).
- Peak flame temperature **closer to adiabatic** flame temperature, much higher thermal NOx production, illustrated in **significantly higher in EI NOx** for turbofan.
- Residence Time: > 6 ms, assumed complete oxidation.
- Max overall pressure ratio (OPR): > 30

Direct operating-point-to-operating-point comparison across the thrust range is thermodynamically unjustified due to order-of-magnitude differences in scale. The sole meaningful correspondence exists at the operating envelope boundaries: "micro-max/macro-idle".

APPENDIX: PHYSICS SURROUNDING COMPARISON

When comparing the micro-turbojet and full-scale turbofan, matching **intensive combustor inlet properties** creates a similar thermodynamic environment for combustion and emissions.

Intensive Properties

Do **not** depend on the amount of mass or the size of the system.

Examples:

- Temperature (T)
- Pressure (p)
- Density (ρ)
- Specific Enthalpy (h)
- Molecular Weight (MW)

Same value regardless of the size or amount of the system.

Key for meaningful comparison between different engines

VS

Extensive Properties

Depend **do** the amount of mass or the size of the system.

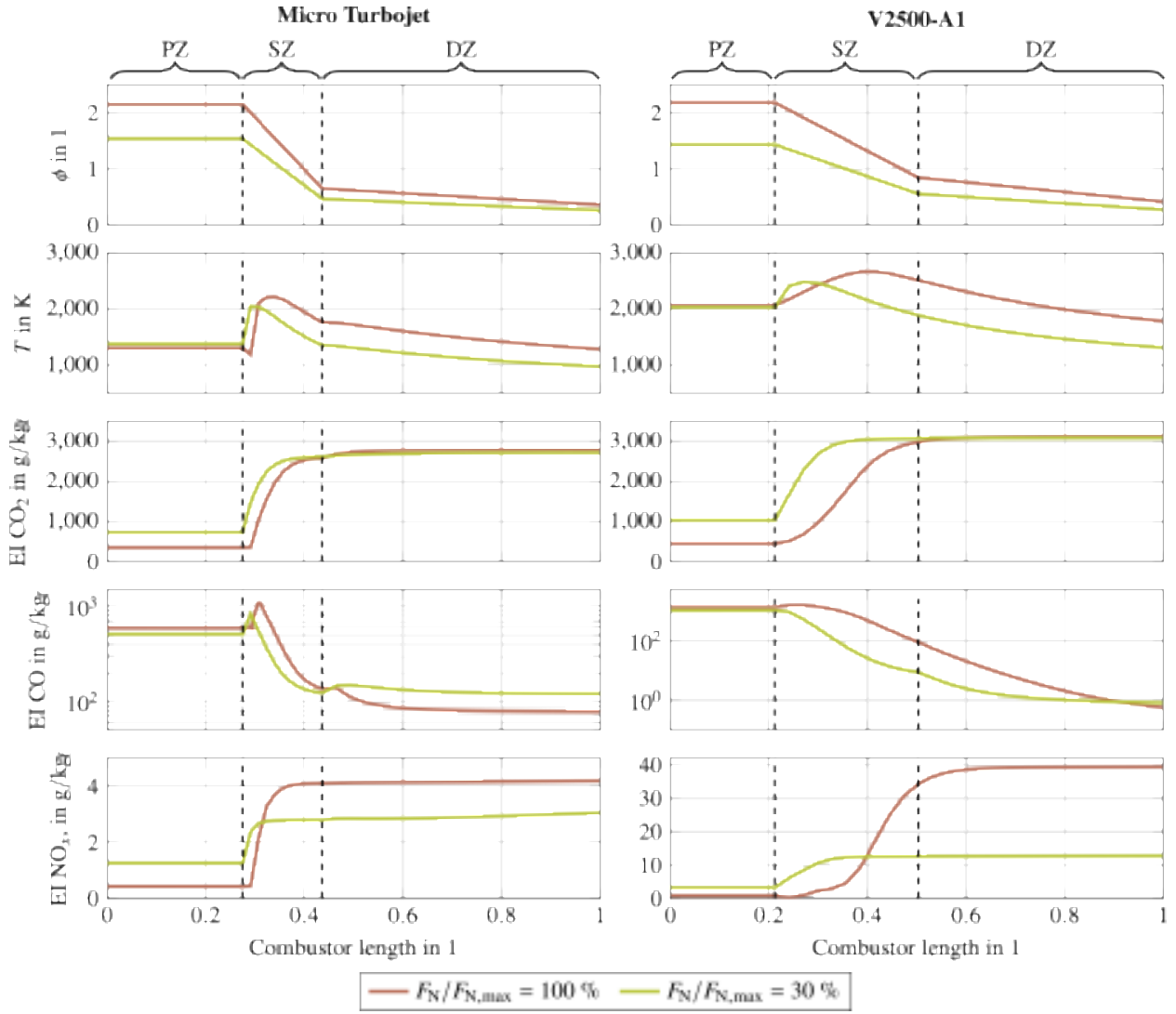
Examples:

- Mass (m)
- Volume (V)
- Total Enthalpy (H)
- Total Energy (E)
- Total Moles (N)

Value changes with the size or amount of the system.

Not suitable for cross-scale engine comparison

APPENDIX: PHYSICS SURROUNDING COMPARISON



APPENDIX: PHYSICS SURROUNDING COMPARISON

	OPR	T_{t3}	P_{t3}
Micro-Turbojet @ max OP	≈ 4	469 K	374 kPa
Full size Turbofan @ idle OP [14]	≈ 3.5-4	453 K	351 kPa

$$k = A \cdot P^n \cdot exp(-\frac{E_a}{RT})$$

k (Reaction Rate Constant): The speed at which the chemical reaction (like soot nucleation) occurs.
A (Pre-exponential Factor): A constant that represents the frequency of molecular collisions.
P (Pressure): The local thermodynamic pressure (374 kPa).
n: An empirical constant showing how sensitive the reaction is to pressure changes.
E_a (Activation Energy): The minimum energy required to kickstart the chemical reaction.
R (Universal Gas Constant): A standard physical constant (8.314 J/(mol·K)).
T (Temperature): The local thermodynamic temperature (469 K).

At the "micro-max/macro-idle" intersection point, the combustor inlet air density, pressure, and temperature of the micro-turbojet and a full-scale turbofan are nearly identical. Combustion chemistry, specifically the formation of emissions that do not equilibrate in a gas turbine, depend entirely on how fast the chemical reactions are progressing (finite rate chemistry). For example, Nox, UHC, CO, Polycyclic Aromatic Hydrocarbons (soot pre-cursors) are driven local conditions and finite rate processes. This is also why a CRN model was used, finite rate chemistry is extremely important to modelling emissions accurately.

APPENDIX: ESTIMATION OF NVPM UNCERTAINTY

$$\eta_j = \eta_{\mathrm{diff},j}\,\eta_{\mathrm{grav},j}\,\eta_{\mathrm{turb},j}\,\eta_{\mathrm{bend},j}.$$

Segment	Description	Re [-]	Model
1	probe to PTD/ejector	394	laminar
2	PTD/ejector block[a]	11237	turbulent
3	main transfer line	2070	laminar[b]
4	branch to 4-way	2201	laminar[b]
5	4-way to CS	1926	laminar[b]
6	CS to SMPS	1926	laminar[b]

$$\eta_{\mathrm{tot,SMPS}}(D_p) = \left[\prod_{j=1}^{6} \eta_j(D_p)\right] \eta_{\mathrm{CS}}(D_p).$$

$$\overline{EI} = \frac{1}{n}\sum_{i=1}^{n} EI_i,$$

$$s_{\mathrm{scan}} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n} EI_i - \overline{EI}^{\,2}}$$

$$u_{\mathrm{rep_mean}} = \frac{s_{\mathrm{scan}}}{\sqrt{n}}.$$

